

Jason Chua Kai Xiang

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Profile: <https://example.com/in/synthetic-student-08>

Skills

C, C++, Arduino, STM32, Python, Git, I2C, SPI

Projects

Plant moisture monitor

Programmed an Arduino microcontroller in C++ to read a soil sensor over I2C, sleep between readings and send data over a serial link. Used interrupts to wake the board on a timer.

Technologies: C++, Arduino, I2C

STM32 motor controller

Wrote a driver in C for an STM32 board to drive a motor through PWM and read a wheel encoder using SPI. Used an oscilloscope and logic analyser to debug timing problems on the real hardware and read the datasheet to set registers.

Technologies: C, STM32, SPI

Small RTOS task scheduler

Ran three tasks on FreeRTOS: sensor sampling, display update and logging, and used queues to share data safely.

Technologies: FreeRTOS, C

2D collision demo

Wrote a small interactive 2D physics demo in C++ where the arrow keys move a ball, using vector maths to detect and resolve collisions with walls.

Technologies: C++

Sensor logger on a single-board computer

Wrote a Python program on a Raspberry Pi that stores sensor readings in a SQLite database with SQL insert statements and exposes the latest readings through a small Flask endpoint.

Technologies: Python, Flask, SQLite

Experience

Robotics club firmware volunteer

Fixed hardware bugs on the club's line-following robot before competitions.

Technologies: C++

Education

BEng Electrical and Computer Engineering - Digital systems and microcontroller modules completed